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# **Module 8 Lecture - Memory**

Introductory Psychology

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# 1 Overview and Introduction

## 1.1 Textbook Learning Objectives

- Discuss the three basic functions of memory
- Describe the three stages of memory storage
- Describe and distinguish between procedural and declarative memory and semantic and episodic memory
- Explain the brain functions involved in memory
- Recognize the roles of the hippocampus, amygdala, and cerebellum
- Compare and contrast the two types of amnesia
- Discuss the unreliability of eyewitness testimony
- Discuss encoding failure
- Discuss the various memory errors
- Compare and contrast the two types of interference

## 1.2 Instructor Learning Objectives

- Make sure you can connect memory back to the all important brain structures!
- Be able to apply the concepts of memory to anecdotal and narrative examples
- Be able to apply an understanding of memory vocabulary to your own life

## 1.3 Introduction

### ! Important

Memory is one of the most crucial cognitive skills we have, and it is stressed constantly in education, because without it, we wouldn't be able to apply our knowledge after we graduate!

- Memory is \_\_\_\_\_
  - When I ask you to \_\_\_\_\_ a certain part of the brain → memory
  - When you \_\_\_\_\_ the name of someone when they walk up to you → memory
  - When you \_\_\_\_\_ about your friends and family → memory
  - “Memory” is even present in small \_\_\_\_\_ things, like tying our shoes

 Discuss: Tell me, what's your earliest memory?

## 2 How Memory Functions

### 2.1 Introduction

- Our **memory** is a \_\_\_\_\_ of encoding information, storing it for some time, and retrieving it from storage
  - The following sections will detail more about each of these parts



 Quick review: which one of these anatomical structures was said to deal a lot with memory during the biopsychology unit?

- ☒ A) Occipital lobe
- ☐ B) Pons
- ☐ C) Hippocampus
- ☐ D) Substantia Nigra

Explanation:

## 2.2 Encoding

- In order to remember something we must first \_\_\_\_\_ it

? If I were to remember something I heard, what sensory receptor would have transduced that information into my brain?

- ☒ A) Taste buds
- ☐ B) Photoreceptors
- ☐ C) Hair cells
- ☐ D) None of the above

Explanation:

- The process of turning that sensory information into \_\_\_\_\_ is called **encoding**
  - There are two types of \_\_\_\_\_ done as part of encoding: automatic and effortful
- As a general rule, the more \_\_\_\_\_ something is, the easier it is to encode

🗣️ Discuss: We've mention how something being 'meaningful' is important in previous units - where and how have we discussed this before?

### 2.2.1 Types of Processing in Encoding

- First, **automatic processing**, is the process by which we automatically encode memories without \_\_\_\_\_ to
  - Tends to be less \_\_\_\_\_ information
  - E.g., what color my car is, what my hair color is, what shirt I wore today and yesterday
- Second, **effortful processing** is that which we do with \_\_\_\_\_ and effort, as the name would imply

- Usually \_\_\_\_\_ complex information
- E.g., memorizing a phone number or an address (in the old days), studying for a midterm exam

 Discuss: I am running to the grocery store, but have no paper or phone to write down a list, so I memorize the vegetables I need and then run out of the house, find my car and leave. What processing did I do to memorize the vegetables and what processing did I do in being able to remember what my car looks like? Why?

### 2.2.2 Types of Encoding

- We can further break down \_\_\_\_\_ of encoding into:
  - \_\_\_\_\_
  - \_\_\_\_\_
  - \_\_\_\_\_
- First of these three, **semantic encoding**
  - Related to \_\_\_\_\_ and the meanings of those words
  - E.g., we remember the word 'elaborate' and it's definition as, "explain something further"
  - We also tend to \_\_\_\_\_ together words in our memory that have similar or related meanings

 What is the most basic 'grouping' unit in our cognition, from the thinking and intelligence unit?

- ☒ A) Schema
- ☐ B) Heuristic
- ☐ C) Algorithm
- ☐ D) Concept

Explanation:

- Next, **visual encoding** is related to our memory of \_\_\_\_\_
  - E.g. I can remember the general visual and diagram of the brain, I visualize the idea of “car” when I see the word
- Finally **acoustic encoding** is related to our ability to encode \_\_\_\_\_ and auditory words
  - E.g., hearing a song and memorizing its melody, learning the 50 states and the alphabet song

### ! Important

All of the above we related to words, but specifically how we interact with those words to memorize them!

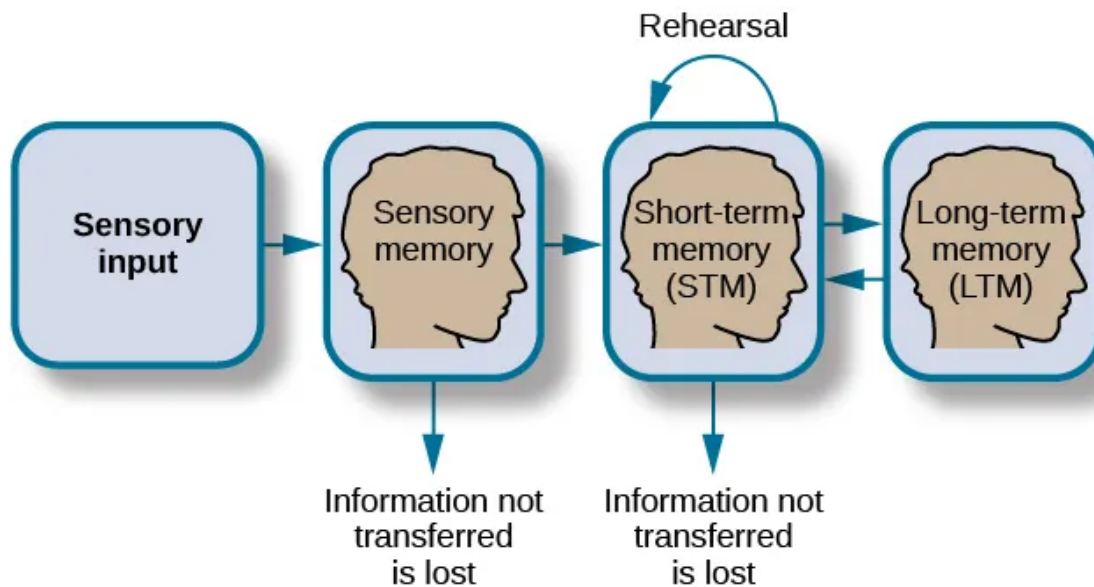
- Interestingly, of the three types above, \_\_\_\_\_ encoding tends to result in the best memory!
  - This is theorized to be related to what we call the **self-reference effect**, which is our tendency to more easily remember information relevant to ourselves

 Discuss: I am listening to the radio in my car on a long drive to a conference, and a popular song is played over and over again. I'm not paying much attention to the music - but as time goes on, I find myself starting to sing along to the chorus each time I hear it. What type of processing is predominantly happening and what type of encoding is happening?

## 2.3 Storage

- If encoding is how we take information \_\_\_\_\_, then **storage** is how we \_\_\_\_\_ it in our mind
  - But, there are multiple \_\_\_\_\_ in a memory becoming \_\_\_\_\_ after being encoded, according to the **Atkinson and Shiffrin Model**:
    - \* \_\_\_\_\_ memory
    - \* \_\_\_\_\_ memory

- \* \_\_\_\_\_ memory
- There are other models of memory \_\_\_\_\_, but we'll focus on this one



**! Important**

We must first properly sense, process, and encode to have a memory in storage!

### 2.3.1 Sensory Memory

- The stage of **sensory memory** is when information is only very briefly held in our memory as sensory information
  - Most sensory memories \_\_\_\_\_ near instantly!
  - Only information which we consider to be \_\_\_\_\_, even if unconsciously, will get retained
  - E.g., the sensory memory of the color of someone's shirt passing on the sidewalk, I will likely not remember past just a moment, *unless* I thought it was a cool shirt



 Discuss: Come up with your own example along the lines above, where a piece of sensory information is likely to get discarded unless you find it useful

### 2.3.2 Short-term Memory

#### Important

'Short-term memory' and 'working memory' often get used similarly, but they are not quite the same.

- The stage of **short-term memory** comes directly after sensory memory
  - Lasts around 15 to 30 \_\_\_\_\_, but still \_\_\_\_\_ than sensory memory
  - Importantly, short-term memory tends to \_\_\_\_\_ information from sensory memory with something that already exists in long term memory.
- We can help move short-term \_\_\_\_\_ to long-term by the process of **rehearsal**, which we might describe in two flavors:
  - \_\_\_\_\_ rehearsal
  - \_\_\_\_\_ rehearsal
- The **active rehearsal**, moves information by simply repeating it \_\_\_\_\_ times
  - E.g., studying the brain by saying the names of the brain over and over again to get the sounds in our head

 Earlier we described several types of encoding - what type of encoding are we doing if we are mostly just focusing on the sounds of the words of the brain?

- ☒ A) Auditory
- ☐ B) Semantic
- ☐ C) Visual
- ☐ D) None of the above

Explanation:

- Another type of rehearsal is **elaborate rehearsal**, which is when we \_\_\_\_\_ new sensory information with that which we already have in long-term memory
  - E.g., I remember processes in our sense/perception module by connecting it closely with what I learned about the brain in the biopsychology module
  - I intentionally try to force a lot of elaborate rehearsal during lecture, to help you move new content to long-term from short term!

 Discuss: Our brains tend to like make all sorts of connections, as it makes us efficient. Explain how the ideas of connections relates to role schemata and event schemata

### 2.3.3 Long-term Memory

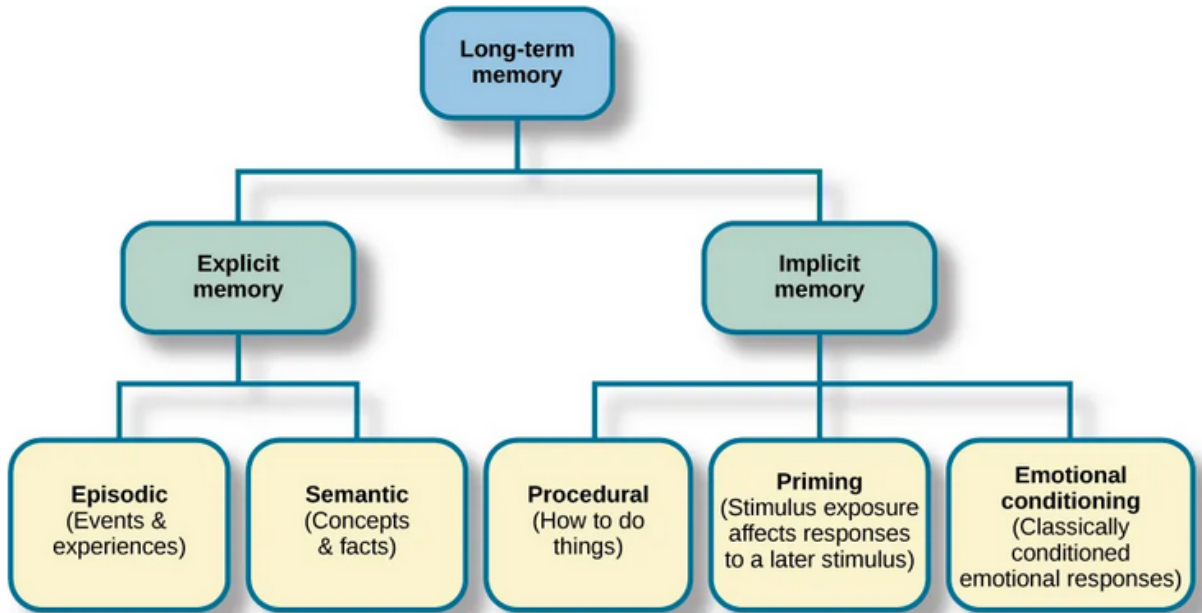
- Our **long-term memory** is where are memories end up after being \_\_\_\_\_ enough or meaningful enough
- As we learned about a bit in our \_\_\_\_\_ unit, our brain connects and \_\_\_\_\_ things to organize
  - When it comes to \_\_\_\_\_, we tend to group words and ideas with **semantic networks**

#### Important

Basically, we were already talking about memory in the last unit, without calling it memory!

- One of the important \_\_\_\_\_ of a semantic network is that we find that as one concept, word or idea is activated - \_\_\_\_\_ ideas also become somewhat activated
  - This creates a sort of \_\_\_\_\_, through our brain to get to certain words and ideas

- The more routes you can take (i.e., having more \_\_\_\_\_), the easier you can remember things!
- Within the broader class of Long-term memory, we have two further \_\_\_\_\_: explicit and implicit, which also have their own divisions



### 2.3.4 Explicit Long-term Memory

- For **explicit memory**, these are those memories that we \_\_\_\_\_ try to maintain, store, and remember in our long-term memory
  - E.g., when I ask you a review question in class, you *explicitly* try to remember it, or when you are trying to remember that really good restaurant you ate at one time
  - This splits into episodic memory and semantic memory
- Then **episodic memory** is all about a \_\_\_\_\_ experience
  - E.g., my memory of teaching last lecture section, my memory of testing patients at work today, your memory of talking with friends in your room earlier in the day

 Discuss: Write your own example of an episode you remember from the last week?

- Finally **semantic memory** deals with knowledge of facts and language

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*"I don't mind not knowing. It doesn't scare me." — Richard P. Feynman*

- E.g., knowing an accurate synonym to the word ‘exhilarating’ is ‘exciting’, being able to correctly name the governor of Michigan, etc.

### 2.3.5 Implicit Long-term Memory

- These **implicit memory** are those which we are not \_\_\_\_\_ or aware of
  - They \_\_\_\_\_ be recalled like explicit memories could be, and are more unseen
  - Further divided into:
    - \* \_\_\_\_\_
    - \* \_\_\_\_\_
    - \* \_\_\_\_\_ conditioning
    - \* We'll just talk about the first two
- Our **procedural memory** are those which tell us \_\_\_\_\_ to do things, like the process involved
  - E.g., Riding a bike, brushing teeth, walking, etc.
  - Procedural memories often start \_\_\_\_\_ during the learning process, but become more automatic with time
- The idea of **priming** happens when we make a \_\_\_\_\_ connection and “primed” or biased to a specific semantic connection
  - E.g., unscramble the following word after this prompt:

For instance, some people really enjoy picnics. They love going into nature, spreading a blanket on the ground, and eating a delicious meal. Now, unscramble the following letters to make a word.

7HETPL

? I am able to 'remember' how to write in cursive and do so without thinking, what type of memory is this best described as?

- ☒ A) Procedural
- ☐ B) Episodic
- ☐ C) Semantic
- ☐ D) Priming

Explanation:

## 2.4 Retrieval

- Once our memories are \_\_\_\_\_ in long-term memory, we obviously hope to be able to pull them out to use! We call this process our **retrieval**.

### ! Important

If we can't retrieve our memories, then they are inaccessible and temporarily useless

- There are 3 \_\_\_\_\_ of retrieval:
  - \_\_\_\_\_
  - \_\_\_\_\_
  - \_\_\_\_\_
- The process of **recall** is when we are able to pull memories from storage with \_\_\_\_\_ cues or hints
  - E.g., I can recall the names of grandparents without any hints, I can recall what my empirical papers are without hints
- Our **recognition** comes when we \_\_\_\_\_ a familiar thing and recognize it
  - E.g., I could identify a dish in a foreign language on a menu as one that I've had before, but couldn't remember the name until I saw the menu

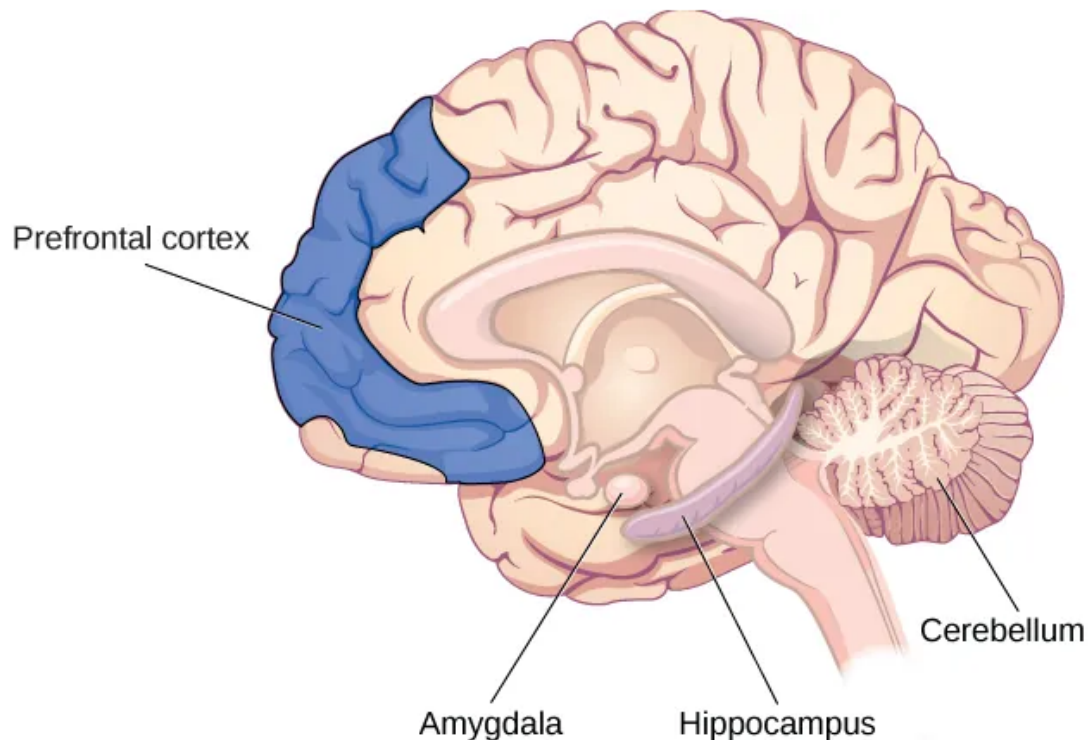
 Discuss: With our understanding of recall and recognition and described above, explain why a short answer test or multiple choice test is more difficult

- Finally, **relearning** is when we just re-encode previously learned information, but now it comes to us faster

### 3 Parts of the Brain Involved with Memory

#### 3.1 Introduction

- Unsurprisingly, the physical structures of the \_\_\_\_\_ have been investigated as having special purposes related to our memories



#### 3.2 The Amygdala

- The amygdala is associated with \_\_\_\_\_ memories, and especially the \_\_\_\_\_ process of emotional memories
  - It also deals with [memory consolidation](#)

##### ! Important

It seems like encoding often works better when an event is emotionally arousing - likely because it is very personally relevant to us

? What was our tendency to remember information relevant to ourself?

- ☒ A) Self-feel effect
- ☐ B) Self-importance effect
- ☐ C) Self-centric effect
- ☐ D) Self-reference effect

Explanation:

### 3.3 The Hippocampus

- Empirical science would suggest that the hippocampus deals much with memory \_\_\_\_\_, as well as \_\_\_\_\_ memory
  - However, function of this area seems mostly related to \_\_\_\_\_ memories, and less to do with implicit types

### 3.4 The Cerebellum and Prefrontal Cortex

- The cerebellum seems fairly connected to \_\_\_\_\_ memory
- The prefrontal cortex seems like it deals much with the initial \_\_\_\_\_ of information

## 4 Conclusion

### 4.1 Recap

- Memory is much more complex than just one simple construct: there are many different types!
- The way we come to store and retrieve information is very important, and there are many parts of encoding and storage process which modify whether sensory information will become long-term memories

- Memory is connected to several structures in the brain, which each have their own unique functions related to different types of memory
- We skipped talking about problems in memory and enhancing memory, for brevity, but consider reading through those sections if you want to learn more!

## Key Terms

### A

**acoustic encoding** Encoding type related to sounds, word in particular [6](#)

**active rehearsal** Type of rehearsal to move information from short-term to long-term memory by repeating information over and over again to better remember it [8](#)

**Atkinson and Shiffrin Model** A model of memory formation that involves the formation of long term memory through sensory input, sensory memory, short-term memory, and then finally long-term memory [6](#)

**automatic processing** When encoding done without any conscious input or intention, memories that automatically form, tends to be less complex information [4](#)

### E

**effortful processing** When encoding done with intention and also tends to be more time-consuming and done on more complex information [4](#)

**elaborate rehearsal** The process of intentionally connecting sensory information to existing long-term information in order to better remember it [9](#)

**encoding** The process by which sensory information is received and initially stored into the brain in some representation [4](#)

**episodic memory** A memory of an experience, narrative, or episode; feels like a story; sometimes described as where, what, when [10](#)

**explicit memory** Memories which we intentional try to store, maintain, and remember [10](#)

### I

**implicit memory** Memories which we are not conscious of or intentional about, cannot be 'recalled' like a fact or an episode [11](#)

### L

**long-term memory** The somewhat infinite/unlimited storage area where memories are kept past being short-term [9](#)

### M

**memory** The process by which we encode, store, and retrieve knowledge and information for some period of time [3](#)

**memory consolidation** Another way of saying new information being moved to long-term memory [13](#)



**P**

**priming** A natural memory that occurs when we expect or readily connect one semantic meaning to another [11](#)

**procedural memory** A memory of how something is done or performed automatically or with little conscious effort [11](#)

**R**

**recall** Accessing memories without any cues or hints; most difficult form of retrieval [12](#)

**recognition** Accessing memories by encountering and identifying it as familiar [12](#)

**rehearsal** A strategy for moving information that you wish to remember for the future from short-term to long-term storage [8](#)

**relearning** Accessing memories by re-practicing and familiarizing yourself, effectively re-encoding the memories [12](#)

**retrieval** The process of pulling a memory from storage for use [12](#)

**S**

**self-reference effect** Tendency for one to remember information relevant to oneself, more so than information with less personal relevance [6](#)

**semantic encoding** Encoding type related to understanding words and their meaning [5](#)

**semantic memory** Memory or knowledge about words, concepts, and language [10](#)

**semantic networks** A network of connections, ideas, and words that are related to one another in our long-term memory [9](#)

**sensory memory** Stage of Atkinson and Shiffrin Model of memory formation that is when something is very momentarily held as sensory information [7](#)

**short-term memory** State of Atkinson and Shiffrin Model of memory formation that is after sensory memory, lasts around 15 to 30 seconds, connects sensory information to existing long-term memories [8](#)

**storage** The process of keeping information in our mind and brain for later retrieval, preceded by retrieval [6](#)

**V**

**visual encoding** Encoding type related to images and visual sense [6](#)